James Madison University

Information Technology (IT) Program

IT 215 Intro to Telecom, Networking & Security

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Lab 2 - Virtual Machines and Network Capture with Wireshark

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19:36:25

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Honor Pledge: I have neither given nor received help on this lab that violates the spirit of the JMU Honor Code.

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Date	*Date*

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1 Introduction

This lab focuses mainly on developing practical skills needed for virtual machines and Wireshark.

2 Lab Exercises : Virtual Machines and Network Capture with Wireshark

2.1 Exercise 1 - Create and Configure a Virtual Machine

T1: Create and Configure a Virtual Machine

Q1: How did you verify the file was created with the correct content? What other ways can be used from the commandline to perform the same verification (list 2 other methods at least)

A1: I verified the file was created with the correct content by using the "cat" command. You can also use a checksum to check the file and going back one level in the directory and check to see if the file exists in there using "ls". See Figure 1.

Figure 1: Verifying Content

T2: Network Connectivity in the VM

Q1: What is your VM's IP address? Can you ping the host machine from your VM? Provide the command and result.

A1: The IP address is 192.168.248.129. You can ping the host machine by using the command "ping" with the prompt to get the host ip address. It will output the host IP and device/network info. See Figure 2.

T3: Install Python

Q1: What commands did you use to install Python 3.13? How did you verify successful installation?

A1: I installed python using "sudo apt install python3". I verified it was successfully installed by running python in the terminal with the command "python3" and having it print the word "hello". See Figure 3.

T4: Clone Your Virtual Machine

```
desktop@desktop-VMware-Virtual-Platform:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:70:25:96 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.248.129/24 brd 192.168.248.255 scope global dynamic noprefixroute ens33
       valid_lft 1338sec preferred_lft 1338sec
    inet6 fe80::20c:29ff:fe70:2596/64 scope link
       valid_lft forever preferred_lft forever
```

Figure 2: IP address

```
desktop@desktop-VMware-Virtual-Platform:~$ sudo apt install python3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2).
python3 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 52 not upgraded.
desktop@desktop-VMware-Virtual-Platform:~$ python3
Python 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> print("hello")
hello
>>>
```

Figure 3: Python3 Installation

Q1: What is the difference between a linked clone and a full clone Summarize your findings in a table with columns for Storage, Speed, and Independence. Use the table below as a template.

A1: A linked clone is directly stemmed from the original VM, it can't exist without the original. A full clone is a copy and paste of the original VM, but runs completely separate. A linked clone takes less storage, loads faster, and needs the original VM to exist. A full clone takes more storage, takes longer to load, but works as its own VM, separate from the original.

T5: Removable Devices and USB Management

Q1: Was Ubuntu VM able to detect and use the USB device? Explain any steps or issues you encountered and how you resolved them.

A1: Yes, Ubuntu was able to detect and access the USB device. While the VM was running and the device was plugged in, I went to "removable devices" and connected to it. It then gave the USB information to the VM opposed to the host. See Figure 4.

T6: VM Power States

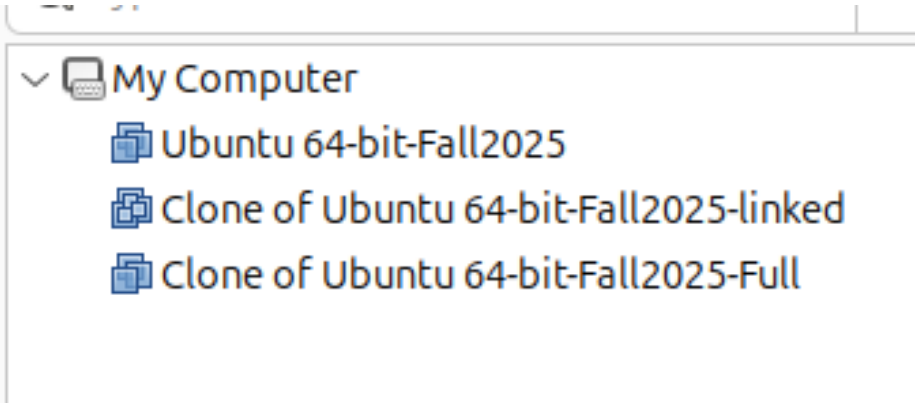


Figure 4: Detecting USB Devices

Q1: What happens to the VM in each case? In a table, compare their effects on saved data and startup/shutdown speed.

A1: When you suspend a VM, that is similar to when you put your computer to sleep. Everything is still present on the computer, just move the mouse and everything is how you left it. Power-off is like unplugging a computer. Anything you've done on it and is not manually saved (like a google window) will be gone and you'll have to boot up the system again to use it.

Dismissal	Saved Data	Startup/Shutdown Time
Suspend	Saves data ready for quick recall	Very Short
Power-Off	Nothing but hard code data saved	Takes Longer

T7: Folder Sharing Between Host and VM

Q1: Describe the steps you used to set up folder sharing. Were you able to read/write files from both the host and the VM?

A1: Getting a shared folder was generally difficult. It involved installing many tools like gedit and vm tools on the VM, but eventually after creating a folder on the host through the VM software settings, then using severl commands in the VM terminal to give the VM access to that folder they had a shared folder that could be written and read by both machines.See Figure 5.

2.2 Exercise 2 - Wireshark Capture and Analysis

T1: Wireshark Setup and Initial Capture

Q1: What does promiscuous mode mean in Wireshark?

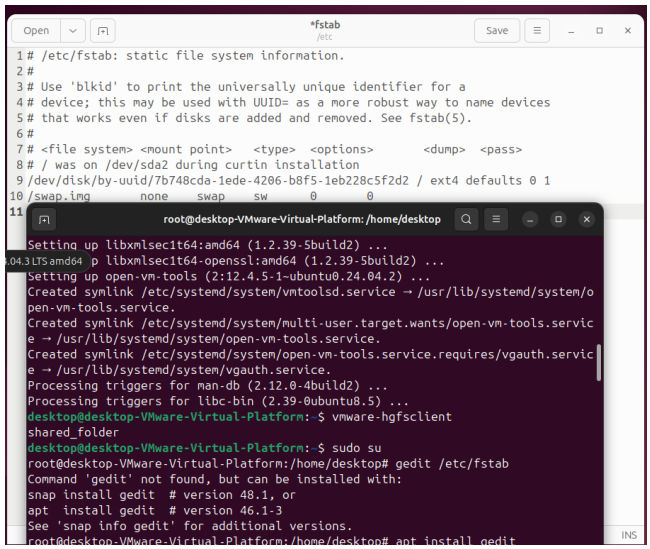


Figure 5: Folder Sharing

A1: The machine will be able to give network information over the connected network, not only the things being directly transmitted through this machine.

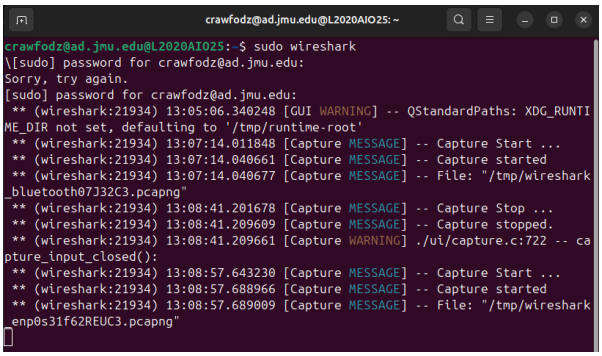
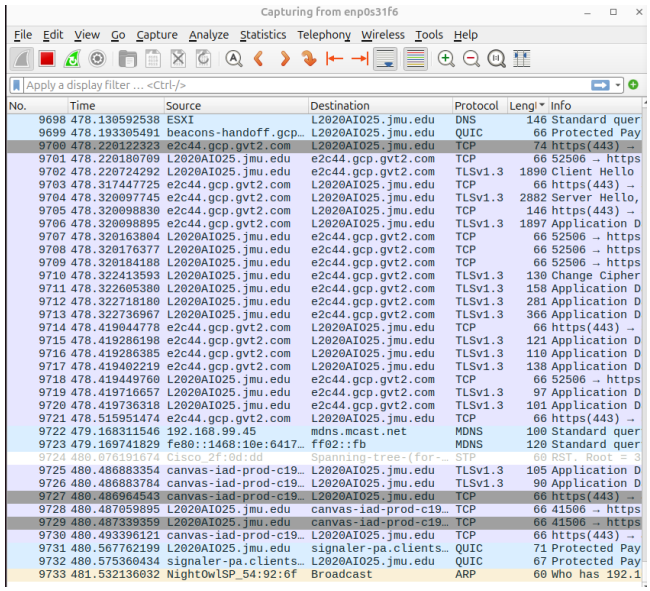


Figure 6: sudo Wireshark Command in Terminal

Q1: Describe the information provided in the top pane and explain the headings of the columns.

A1: The information in the top pane is all the different packets over the internet I'm connected to. "No." is the number of the packet since the start of the capture. Time is the time the packet was received since the capture (in seconds). Source is where the packet

came from. Destination is the IP address of where the information is going. Protocol is telling you which protocol was used for that connection. The length is the length of the packets. Info is general information about the packet. See Figure 7.



No.	Time	Source	Destination	Protocol	Lengt	Info
9698	478.130592538	ESXI	L2020AIO25.jmu.edu	DNS	146	Standard quer
9699	478.193305491	beacons-handoff.gcp…	L2020AIO25.jmu.edu	QUIC	66	Protected Pay
9700	478.220122323	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TCP	74	https(443) →
9701	478.220180709	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TCP	66	52506 → https
9702	478.220724292	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	1890	Client Hello
9703	478.317447725	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TCP	66	https(443) →
9704	478.320097745	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TLSv1.3	2882	Server Hello,
9705	478.320098830	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TCP	146	https(443) →
9706	478.320098895	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TLSv1.3	1897	Application D
9707	478.320163804	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TCP	66	52506 → https
9708	478.320176377	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TCP	66	52506 → https
9709	478.320184188	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TCP	66	52506 → https
9710	478.322413593	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	130	Change Cipher
9711	478.322605380	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	158	Application D
9712	478.322718180	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	281	Application D
9713	478.322736967	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	366	Application D
9714	478.419044778	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TCP	66	https(443) →
9715	478.419286198	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TLSv1.3	121	Application D
9716	478.419286385	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TLSv1.3	110	Application D
9717	478.419402219	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TLSv1.3	138	Application D
9718	478.419449760	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TCP	66	52506 → https
9719	478.419716657	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	97	Application D
9720	478.419736318	L2020AIO25.jmu.edu	e2c44.gcp.gvt2.com	TLSv1.3	101	Application D
9721	478.515951474	e2c44.gcp.gvt2.com	L2020AIO25.jmu.edu	TCP	66	https(443) →
9722	479.168311546	192.168.99.45	mdns.mcast.net	MDNS	100	Standard quer
9723	479.169741829	fe80::1468:10e:6417…	ff02::fb	MDNS	120	Standard quer
9724	480.976191674	Cisco_2f:9d:dd	Spanning-tree-(for-…	STP	60	RST. Root = 3
9725	480.486883354	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	105	Application D
9726	480.486883784	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	90	Application D
9727	480.486964543	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) →
9728	480.487059895	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	41506 → https
9729	480.487339359	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	41506 → https
9730	480.493396121	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) →
9731	480.567762199	L2020AIO25.jmu.edu	signaler-pa.clients…	QUIC	71	Protected Pay
9732	480.575360434	signaler-pa.clients…	L2020AIO25.jmu.edu	QUIC	67	Protected Pay
9733	481.532136032	NightOwlSP_54:92:6f	Broadcast	ARP	60	Who has 192.1

Figure 7: Wireshark Panels

Q1: Provide and example after sorting by the Protocol column.

A1: See Figure 8.

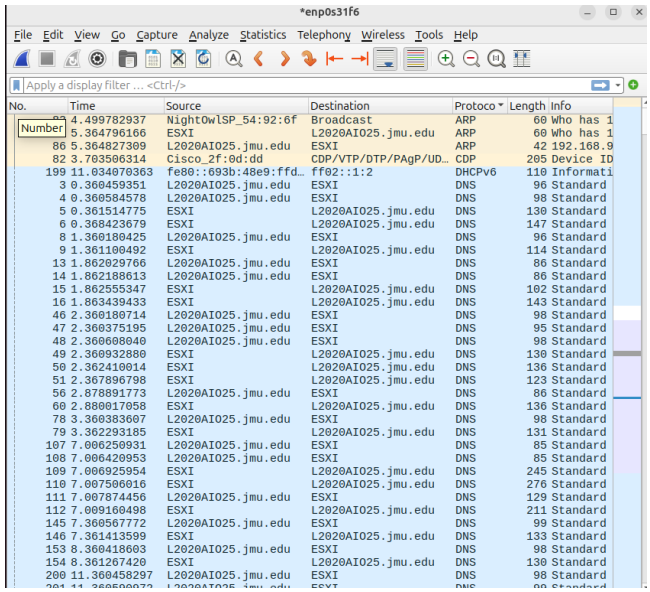
Q1: How can you change the bottom pane view from Hex View to Bits? How would you convert the decimal number 87 to Hex and Binary representations?

A1: You can change the bottom pane from hex to bits by right clicking on the pane, and clicking the pits option to display the data that way. You would convert the decimal number 87 to binary and hexadecimal by doing the math to convert it, whether that be by hand or through a converter. 87 in binary is 1010111 and in hexadecimal is 57

T2: Capturing Frames and Reviewing Protocols

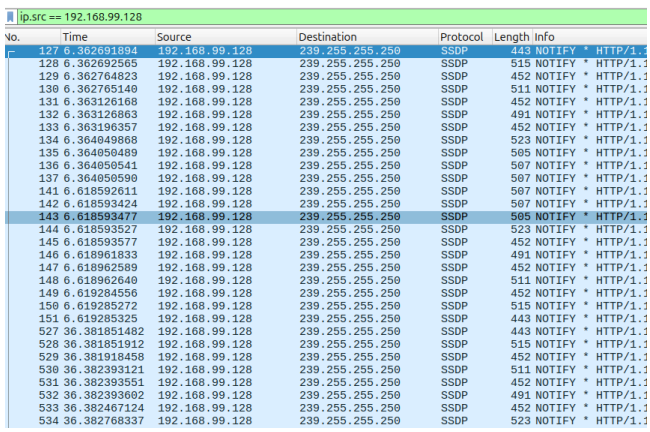
Q1: Compare the terminal output of nslookup and the content of the corresponding DNS response captured by Wireshark.

A1: See Figure 9 and Figure 10.



No.	Time	Source	Destination	Protoco	Length	Info
82	4.499782937	NightOwlSP_54:92:6f	Broadcast	ARP	60	Who has 1
83	5.364796166	ESXI	L2020AIO25.jmu.edu	ARP	60	Who has 1
86	5.364827309	L2020AIO25.jmu.edu	ESXI	ARP	42	192.168.9
82	3.703506314	Cisco_2f:0d:dd	CDP/VTP/DTP/PAgP/UD…	CDP	205	Device ID
199	11.034070363	fe80::693b:48e9:ffd…	ff02::1:2	DHCPv6	110	Informati
3	0.360459351	L2020AIO25.jmu.edu	ESXI	DNS	96	Standard
4	0.360584578	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard
5	0.361514775	ESXI	L2020AIO25.jmu.edu	DNS	130	Standard
6	0.368423679	ESXI	L2020AIO25.jmu.edu	DNS	147	Standard
8	1.360180425	L2020AIO25.jmu.edu	ESXI	DNS	96	Standard
9	1.361100492	ESXI	L2020AIO25.jmu.edu	DNS	114	Standard
13	1.862029766	L2020AIO25.jmu.edu	ESXI	DNS	86	Standard
14	1.862188613	L2020AIO25.jmu.edu	ESXI	DNS	86	Standard
15	1.862555347	ESXI	L2020AIO25.jmu.edu	DNS	102	Standard
16	1.863439433	ESXI	L2020AIO25.jmu.edu	DNS	143	Standard
46	2.360180714	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard
47	2.360375195	L2020AIO25.jmu.edu	ESXI	DNS	95	Standard
48	2.360608040	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard
49	2.360932880	ESXI	L2020AIO25.jmu.edu	DNS	130	Standard
50	2.362410014	ESXI	L2020AIO25.jmu.edu	DNS	136	Standard
51	2.367896798	ESXI	L2020AIO25.jmu.edu	DNS	123	Standard
56	2.878891773	L2020AIO25.jmu.edu	ESXI	DNS	86	Standard
60	2.880017058	ESXI	L2020AIO25.jmu.edu	DNS	136	Standard
78	3.360383607	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard
79	3.362293185	ESXI	L2020AIO25.jmu.edu	DNS	131	Standard
107	7.006250931	L2020AIO25.jmu.edu	ESXI	DNS	85	Standard
108	7.006420953	L2020AIO25.jmu.edu	ESXI	DNS	85	Standard
109	7.006925954	ESXI	L2020AIO25.jmu.edu	DNS	245	Standard
110	7.007506016	ESXI	L2020AIO25.jmu.edu	DNS	276	Standard
111	7.007874456	L2020AIO25.jmu.edu	ESXI	DNS	129	Standard
112	7.009160498	ESXI	L2020AIO25.jmu.edu	DNS	211	Standard
145	7.360567772	L2020AIO25.jmu.edu	ESXI	DNS	99	Standard
146	7.361413599	ESXI	L2020AIO25.jmu.edu	DNS	133	Standard
153	8.360418603	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard
154	8.361267420	ESXI	L2020AIO25.jmu.edu	DNS	130	Standard
200	11.360458297	L2020AIO25.jmu.edu	ESXI	DNS	98	Standard

Figure 8: Sorted Protocol Panel



ip.src == 192.168.99.128

No.	Time	Source	Destination	Protocol	Length	Info
127	6.362691894	192.168.99.128	239.255.255.250	SSDP	443	NOTIFY * HTTP/1.1
128	6.362692565	192.168.99.128	239.255.255.250	SSDP	515	NOTIFY * HTTP/1.1
129	6.362764823	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
130	6.362765140	192.168.99.128	239.255.255.250	SSDP	511	NOTIFY * HTTP/1.1
131	6.363126168	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
132	6.363126863	192.168.99.128	239.255.255.250	SSDP	491	NOTIFY * HTTP/1.1
133	6.363196357	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
134	6.364049868	192.168.99.128	239.255.255.250	SSDP	523	NOTIFY * HTTP/1.1
135	6.364050489	192.168.99.128	239.255.255.250	SSDP	505	NOTIFY * HTTP/1.1
136	6.364050541	192.168.99.128	239.255.255.250	SSDP	507	NOTIFY * HTTP/1.1
137	6.364050590	192.168.99.128	239.255.255.250	SSDP	507	NOTIFY * HTTP/1.1
141	6.618592611	192.168.99.128	239.255.255.250	SSDP	507	NOTIFY * HTTP/1.1
142	6.618593424	192.168.99.128	239.255.255.250	SSDP	507	NOTIFY * HTTP/1.1
143	6.618593477	192.168.99.128	239.255.255.250	SSDP	505	NOTIFY * HTTP/1.1
144	6.618593527	192.168.99.128	239.255.255.250	SSDP	523	NOTIFY * HTTP/1.1
145	6.618593577	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
146	6.618961833	192.168.99.128	239.255.255.250	SSDP	491	NOTIFY * HTTP/1.1
147	6.618962589	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
148	6.618962640	192.168.99.128	239.255.255.250	SSDP	511	NOTIFY * HTTP/1.1
149	6.619284556	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
150	6.619285272	192.168.99.128	239.255.255.250	SSDP	515	NOTIFY * HTTP/1.1
151	6.619285325	192.168.99.128	239.255.255.250	SSDP	443	NOTIFY * HTTP/1.1
527	36.381851482	192.168.99.128	239.255.255.250	SSDP	443	NOTIFY * HTTP/1.1
528	36.381851912	192.168.99.128	239.255.255.250	SSDP	515	NOTIFY * HTTP/1.1
529	36.381918458	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
530	36.382393121	192.168.99.128	239.255.255.250	SSDP	511	NOTIFY * HTTP/1.1
531	36.382393551	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
532	36.382393602	192.168.99.128	239.255.255.250	SSDP	491	NOTIFY * HTTP/1.1
533	36.382467124	192.168.99.128	239.255.255.250	SSDP	452	NOTIFY * HTTP/1.1
534	36.382768337	192.168.99.128	239.255.255.250	SSDP	523	NOTIFY * HTTP/1.1

Figure 9: ip.src

Q1: Attach a copy of your .pcap file to your report as specified in the lab deliverables. Note: Make sure that you save the file and .pcap and NOT .pcapng. Failure to do so will make you lose points for this part

A1: See Canvas Attachment

Q1: Present the binary encoding for "A", "3", and carriage return. What are the hexadecimal (Hex) codes for "G", "E", and "T"?

tcp && !http

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	L2020AIO25.jmu.edu	syslog.jmu.edu	RSH	208	Client -> Server
2	0.000104324	L2020AIO25.jmu.edu	exabeam.jmu.edu	RSH	208	Client -> Server
3	0.000810262	exabeam.jmu.edu	L2020AIO25.jmu.edu	TCP	66	shell(514) → 3879
4	0.000810467	syslog.jmu.edu	L2020AIO25.jmu.edu	TCP	66	shell(514) → 3955
26	1.421524213	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	74	36070 → https(443
27	1.427212941	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	74	https(443) → 3607
28	1.427277433	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	36070 → https(443
29	1.427749620	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	1975	Client Hello (SNI
30	1.433429461	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) → 3607
31	1.433429896	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) → 3607
32	1.433849247	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	300	Server Hello, Cha
33	1.433877765	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	36070 → https(443
34	1.438527105	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	130	Change Cipher Spe
35	1.438673591	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	158	Application Data
36	1.438909065	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	1382	Application Data
37	1.439345303	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	790	Application Data,
38	1.444127530	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) → 3607
39	1.444127950	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	245	Application Data
40	1.444128002	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	137	Application Data
41	1.444204107	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	36070 → https(443
42	1.444256154	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TLSv1.3	97	Application Data
43	1.444750745	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) → 3607
45	1.492806969	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TCP	66	https(443) → 3607
46	1.493338813	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	2069	Application Data
47	1.493339201	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	97	Application Data
48	1.493444567	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	36070 → https(443
58	1.502061850	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	1756	Application Data
59	1.502062286	canvas-iad-prod-c19…	L2020AIO25.jmu.edu	TLSv1.3	97	Application Data
60	1.502147917	L2020AIO25.jmu.edu	canvas-iad-prod-c19…	TCP	66	36070 → https(443
93	3.313778992	L2020AIO25.jmu.edu	80.54.36.34.bc.goog…	TLSv1.2	105	Application Data

Figure 10: Filtering for TCP not HTTP

A1: The binary representation for A, 3, and carriage return are 01000001, 11, 00001101. The Hex codes for G, E and T are 47, 45, 54. See Figure 11.

```
0000  10101000 00111100 10100101 00100
0008  11100000 10011010 01010100 11111
0010  00000000 01101101 10110000 01010
0018  10000001 11111111 11000000 10101
0020  01100011 11010111 00000000 00110
0028  01101110 11000011 01000010 01011
0030  00000000 00000001 00000000 00000
0038  00110101 00110001 00000001 00110
0040  00110010 00110100 00000111 01101
0048  01100100 01110010 00000100 01100
0050  00000000 00001100 00000000 00000
0058  00000000 00000001 00000000 00000
```

Figure 11: Binary Encoding

T3: Display Filtering in Wireshark

Q1: Why are some frames labeled as HTTP under the protocol heading after applying the TCP filter?

A1: Some of the HTTP frames use TCP in the second layer, meaning they fall under the TCP category even if they are listed as HTTP

in wireshark.

Q1: Demonstrate using filter syntax to exclude HTTP while retaining TCP. Give an example of filtering for packets from a specific source IP address, with a screenshot.

A1: See Figure 12.

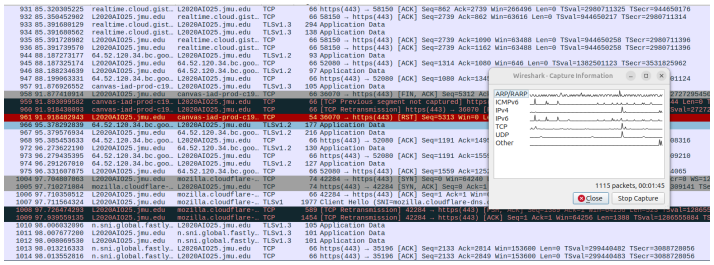


Figure 12: Filter Syntax

Q1: Describe what happened after applying the ip.ttl filter. Can you filter for specific source and destination IP addresses at the same time? Provide evidence.

A1: After applying the ip.ttl filter there was nothing. After a search, google told me that ip 128 is pinging windows devices, and as there are none in this room connected to the same access point, that makes sense. You can filter source and destination ip addresses at the same time by using "&&" in between the two filters. See Figure 13 and Figure 14.

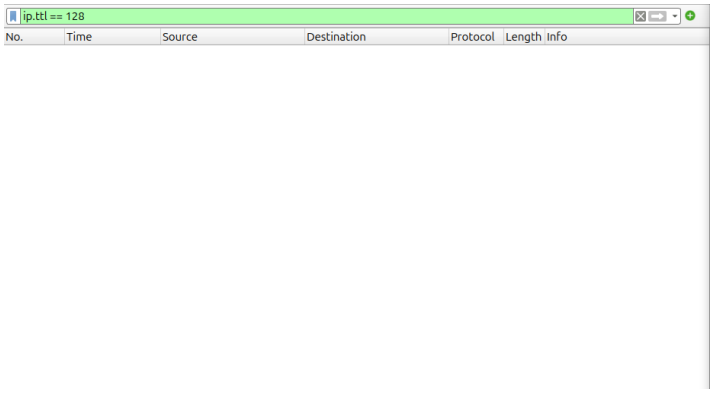


Figure 13: ip.ttl Filter

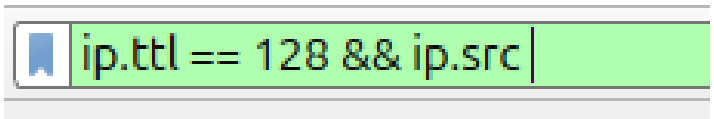


Figure 14: Filtering Source and Destination IP Addresses

2.3 Exercise 3 - Wireshark Capture and TCP/IP Layer Matching

T1: Capturing Data from a New Website

Q1: How did you identify traffic between your machine and the website (e.g., using your public IP address)? Did you successfully filter for this session? Attach a screenshot

A1: I was not able to successfully filter this session because I was unable to find the pings between my machine and CNN (the website I chose). BUT, you are able to filter through a specific application as long as you have the correct IP address. See Figure 15.

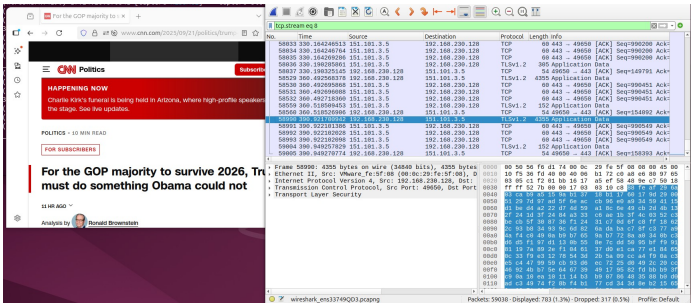


Figure 15: CNN Website

T2: Drawing and Matching the 5-Layer Network Model

Q1: Show your drawing or mapping. Which protocol(s) matched each layer in your example packet?

A1: See Figure 16.

T3: Identifying Layer-Specific Features in Wireshark

Q1: Provide screenshots and answers for each of the four features above (MAC, IP, Port, Application). Describe how you followed the TCP stream and what you learned.

A1: Mac address - 00:0c:29:fe:5f:08 (Data Link layer, IP (Network layer), Source IP - 192.168.239.128, Destination IP - 34.107.221.82, Port Numbers - Port Numbers (Transport layer) - 52560, 80, Application Identification (Application Layer) - shows the full URL

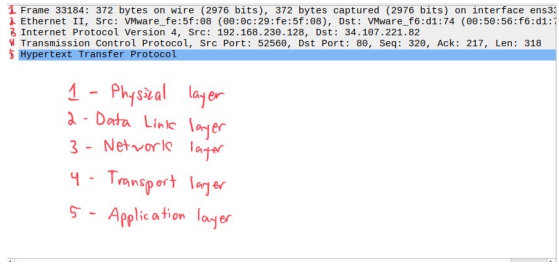


Figure 16: 5 Layer Network Model

from the requested application (in this case the packet was for Firefox). The TCP follow shows the accepted connections and then the connected information between the host and the connected application using the drop down menu for that layer in Wireshark.See Figure 17.

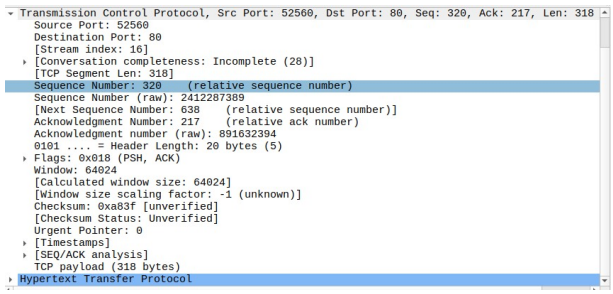


Figure 17: TCP Stream

3 Conclusion

3.1 Summary

Overall, this lab strengthened my technical skills in virtualization as well as in network analysis. I explored device connections such as USB drives and shared folders and used different VM power states and cloning methods.

3.2 Expectancy-Value-Cost

Expectancy
I felt pretty confident in completing the lab because of my prior exposure to Linux commands. Some steps like checking IP addresses were pretty straightforward but configuring share folders and cloning VMs were slightly more challenging.
Value
I think I gained valuable experience from this lab as I was able to connect more theoretical concepts about TCP and IP models to network traffic. Using Wireshark made protocols like DNS, HTTP, TCP more concrete.
Cost
We put in around 4 hours of work on this lab.

3.3 Suggestions

Instruction - Do not include the question in your lab report.
Overall it was a straightforward lab, although some questions were worded vaguely eg.) installing Scratch.

3.4 Best Practices

Some best practices would be to read the questions carefully to fully understand what the problem is asking, and document step by step procedures.

4 Lab References

[1] OpenAI, *Response from ChatGPT: Wireshark Help.* Accessed: Sept. 22, 2025. Available: https://chat.openai.com/

[2] OpenAI, *Response from ChatGPT: Overleaf format help.* Accessed: Sept. 21, 2025. Available: https://chat.openai.com/

5 Acknowledgments

There is no one to acknowledge for this specific lab.